

Online Appendix for TPDL 2025 submission titled: Data Discovery using LLMs - A study of data user behaviour

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Table 1. Number of sessions (|s|) and percentage of queries/responses (%) in which dialogue acts were present for prompts issued by participants and responses from LLMs for unguided (U) and guided (G) tasks. Percentages are rounded on 2 digits. Note that the maximum number of sessions per actor and LLM is 16 in the upper two settings, while the maximum number of sessions is 32 in the combined setting.

LLM Actor Task	ChatGPT							
	Participant				LLM			
	s	%	s	%	s	%	s	%
Statement	11	37.29%	15	43.1%	16	100.0%	16	98.28%
Opinion	0	0.0%	4	6.9%	1	1.69%	2	6.9%
Question	14	52.54%	16	70.69%	11	61.02%	13	68.97%
Answers	1	5.08%	5	12.07%	5	22.03%	10	24.14%
Appreciation	0	0.0%	2	3.45%	1	1.69%	5	13.79%
Apology	0	0.0%	1	1.72%	0	0.0%	0	0.0%
Thanking	2	3.39%	6	10.34%	1	1.69%	0	0.0%
Action	13	45.76%	12	34.48%	0	0.0%	0	0.0%
Offer	0	0.0%	0	0.0%	15	84.75%	16	100.0%
Opening	2	5.08%	9	15.52%	0	0.0%	8	17.24%
Closing	0	0.0%	1	1.72%	0	0.0%	1	1.72%
Hold	0	0.0%	0	0.0%	1	1.69%	4	6.9%

LLM Actor Task	Perplexity							
	Participant				LLM			
	s	%	s	%	s	%	s	%
Statement	9	27.94%	14	34.18%	16	98.53%	16	97.47%
Opinion	1	1.47%	1	1.27%	0	0.0%	1	1.27%
Question	14	58.82%	16	81.01%	1	1.47%	4	8.86%
Answers	2	2.94%	0	1.27%	11	23.53%	13	31.65%
Appreciation	0	0.0%	0	0.0%	0	0.0%	8	11.39%
Apology	0	0.0%	0	0.0%	2	5.88%	5	6.33%
Thanking	0	0.0%	4	7.59%	0	0.0%	1	1.27%
Action	13	52.94%	10	21.52%	0	0.0%	0	0.0%
Offer	0	0.0%	0	0.0%	2	2.94%	9	22.78%
Opening	0	0.0%	12	16.46%	0	0.0%	14	20.25%
Closing	0	0.0%	1	1.27%	0	0.0%	0	0.0%
Hold	0	0.0%	0	0.0%	0	0.0%	0	0.0%

LLM Actor Task	ChatGPT + Perplexity Combined							
	Participant				LLM			
	s	%	s	%	s	%	s	%
Statement	20	32.28%	29	37.96%	32	99.21%	32	97.81%
Opinion	1	0.79%	5	3.65%	1	0.79%	3	3.65%
Question	28	55.91%	32	76.64%	12	29.13%	17	34.31%
Answers	3	3.94%	5	5.84%	16	22.83%	23	28.47%
Appreciation	0	0.0%	2	1.46%	1	0.79%	13	12.41%
Apology	0	0.0%	1	0.73%	2	3.15%	5	3.65%
Thanking	2	1.57%	10	8.76%	1	0.79%	1	0.73%
Action	26	49.61%	22	27.01%	0	0.0%	0	0.0%
Offer	0	0.0%	0	0.0%	17	40.94%	25	55.47%
Opening	2	2.36%	21	16.06%	0	0.0%	22	18.98%
Closing	0	0.0%	2	1.46%	0	0.0%	1	0.73%
Hold	0	0.0%	0	0.0%	1	0.79%	4	2.92%

Table 2. Session characteristics for the unguided and guided tasks by different experience groups and Wilcoxon rank-sum test between groups.

	low	Unguided high	Wilcoxon rank- sum test	low	Guided high	Wilcoxon rank- sum test
Experience in data search						
Average number of queries	3.8182 (0.5692)	4.0476 (0.4704)	0.2020 (0.8658)	3.5455 (0.6923)	4.6667 (0.6108)	-1.2870 (0.2060)
Average query length (words)	18.5820 (3.5004)	23.4343 (3.2571)	-0.8730 (0.3941)	34.0955 (3.4282)	33.2425 (3.3220)	0.2980 (0.7846)
Average response length (words)	286.1675 (33.4878)	312.6077 (28.8415)	-0.2180 (0.8452)	196.0258 (18.7828)	213.3322 (12.2508)	-1.1310 (0.2713)
Average noun to total words ratio	0.3637 (0.0232)	0.3593 (0.0200)	0.5750 (0.5842)	0.3081 (0.0194)	0.2844 (0.0010)	0.8530 (0.4107)
Average number of emojis	2.8957 (2.0973)	0.9421 (0.3757)	0.6620 (0.5216)	0.7182 (0.4010)	1.8044 (0.7624)	-0.3650 (0.7341)
N	11	21		11	21	
Experience in working with data						
Average number of queries	3.4706 (0.5431)	4.5333 (0.4350)	-2.135* (0.0334)	3.7647 (0.6616)	4.8667 (0.6537)	-1.6460 (0.1023)
Average query length (words)	26.5354 (3.9730)	16.3614 (2.0178)	1.870 (0.0623)	36.9839 (3.1839)	29.6278 (3.6151)	1.4920 (0.1425)
Average response length (words)	320.4142 (36.3643)	284.3710 (22.6506)	0.736 (0.4784)	209.4742 (14.9163)	205.0132 (14.3637)	-0.1700 (0.8816)
Average noun to total words ratio	0.3440 (0.0178)	0.3798 (0.0250)	-0.736 (0.4784)	0.3011 (0.0151)	0.2828 (0.0103)	0.9250 (0.3699)
Average number of emojis	2.3775 (1.3816)	0.7479 (0.3804)	0.6730 (0.5061)	2.0500 (0.9126)	0.7294 (0.3774)	1.1290 (0.2758)
N	17	15		17	15	
Experience with LLMs						
Average number of queries	5.3333 (0.9129)	3.4348 (0.3004)	1.9860 (0.0530)	6.1111 (0.8731)	3.5652 (0.4903)	2.7830** (0.0042)
Average query length (words)	13.4645 (2.5827)	25.0149 (3.0250)	-2.494* (0.0111)	26.7232 (5.2672)	36.2015 (2.5747)	-1.9070 (0.0583)
Average response length (words)	270.2540 (35.9431)	316.5356 (27.0741)	-0.8590 (0.4088)	219.1525 (24.9455)	202.7777 (10.6568)	0.6080 (0.5640)
Average noun to total words ratio	0.3748 (0.0290)	0.3553 (0.0180)	0.4820 (0.6506)	0.2844 (0.0180)	0.2957 (0.0112)	-0.1890 (0.8693)
Average number of emojis	0.4984 (0.2153)	2.0500 (1.0443)	-0.1450 (0.9279)	2.0306 (1.3761)	1.1964 (0.5058)	0.0240 (0.9632)
N	9	23		9	23	

Notes. Mean with standard error for two groups (high and low) of experience each in searching for data, working with data, and using LLMs, z-statistic and exact p-value for Wilcoxon rank-sum test performed between two groups, number of observations in each experience group, * $p < 0.05$, ** $p < 0.005$.

Table 3. Participants’ user experience (five UEQ scales) for the unguided and guided tasks by different experience groups and Wilcoxon rank-sum test between groups.

	low	Unguided high	Wilcoxon rank- sum test	low	Guided high	Wilcoxon rank- sum test
Experience in data search						
Attractiveness	1.1970 (0.3615)	0.9083 (0.2178)	0.9720 (0.3421)	1.6212 (0.3610)	0.8750 (0.2313)	1.9030 (0.0578)
Efficiency	1.7045 (0.2672)	1.0000 (0.2440)	1.9720* (0.0485)	1.8636 (0.2887)	0.8250 (0.2739)	2.5240* (0.0102)
Perspicuity	2.0682 (0.1910)	1.6190 (0.1781)	1.496 (0.1387)	1.8409 (0.3238)	1.4643 (0.1820)	1.3960 (0.1684)
Dependability	0.9167 (0.2700)	0.7763 (0.1528)	0.3720 (0.7239)	1.1000 (0.3480)	0.7632 (0.2081)	0.7610 (0.4604)
Stimulation	1.1364 (0.4400)	0.9881 (0.1932)	0.8370 (0.4142)	1.7045 (0.3468)	0.7738 (0.2633)	1.972* (0.0486)
N	11	21		11	21	
Experience in working with data						
Attractiveness	0.9412 (0.2734)	1.0952 (0.2619)	-0.2780 (0.7915)	1.4216 (0.2569)	0.7976 (0.3116)	1.4120 (0.1633)
Efficiency	1.2500 (0.2720)	1.2500 (0.2761)	0.2390 (0.8210)	1.4559 (0.2628)	0.8750 (0.3629)	1.2930 (0.2026)
Perspicuity	1.7941 (0.1991)	1.7500 (0.1952)	0.0190 (0.9925)	1.6912 (0.2344)	1.4833 (0.2308)	0.4740 (0.6466)
Dependability	0.8500 (0.1669)	0.7885 (0.2190)	0.3250 (0.7574)	1.0469 (0.2144)	0.6731 (0.3041)	1.2560 (0.2166)
Stimulation	1.0294 (0.2736)	1.0500 (0.2826)	-0.0190 (0.9926)	1.5000 (0.2590)	0.6333 (0.3406)	1.839 (0.0668)
N	17	15		17	15	
Experience with LLMs						
Attractiveness	0.6250 (0.4270)	1.1449 (0.2046)	-1.1760 (0.2499)	0.1875 (0.3719)	1.4710 (0.2042)	-2.6910* (0.0055)
Efficiency	0.8438 (0.4793)	1.3913 (0.1956)	-1.0210 (0.3193)	0.1250 (0.4864)	1.5652 (0.1977)	-2.6020* (0.0076)
Perspicuity	1.5278 (0.3266)	1.8696 (0.1431)	-0.7380 (0.4741)	1.0278 (0.2840)	1.8152 (0.1814)	-2.0850* (0.0364)
Dependability	0.3750 (0.3339)	0.9432 (0.1356)	-1.4680 (0.1497)	-0.0834 (0.3689)	1.1304 (0.1742)	-2.7060** (0.0046)
Stimulation	0.8611 (0.3310)	1.1087 (0.2388)	-1.116 (0.2746)	0.1389 (0.4392)	1.4674 (0.2145)	-2.5040* (0.0108)
N	9	23		9	23	

Notes. Mean with standard error for five UEQ scales [2] for two groups (high and low) of experience each in searching for data, working with data, and using LLMs, z-statistic and exact p-value for Wilcoxon rank-sum test performed between two groups, number of observations in each experience group, UEQ scales range from -3 to +3, * $p < 0.05$, ** $p < 0.005$.

Table 4. Participants' user experience (NASA Task Load Index) for the unguided and guided tasks by different experience groups and Wilcoxon rank-sum test between groups.

	Unguided			Guided		
	low	high	Wilcoxon rank- sum test	low	high	Wilcoxon rank- sum test
Experience in data search						
Mental demand	3.6364 (0.3636)	3.5714 (0.3623)	0.2130 (0.8464)	3.9091 (0.4946)	3.2857 (0.3247)	1.0690 (0.3018)
Physical demand	2.0000 (0.3814)	1.5238 (0.2140)	-0.1070 (0.9804)	2.3636 (0.4912)	1.6667 (0.2423)	1.1900 (0.2556)
Temporal demand	3.3636 (0.4527)	3.8571 (0.3330)	-2.4260* (0.0141)	2.8182 (0.4635)	3.2857 (0.3725)	-0.7700 (0.4549)
Performance	3.4545 (0.5110)	4.0000 (0.4082)	-1.1190 (0.2691)	3.0909 (0.5301)	3.7143 (0.3460)	-1.1590 (0.2583)
Effort	3.4545 (0.4341)	3.3333 (0.2954)	1.2720 (0.2082)	3.0000 (0.3568)	2.9048 (0.2752)	0.2870 (0.8006)
Frustration	2.8182 (0.5363)	3.0000 (0.3842)	1.1310 (0.2731)	2.7273 (0.5409)	3.0476 (0.3682)	-0.5340 (0.6208)
N	11	21		11	21	
Experience in working with data						
Mental demand	3.6471 (0.3529)	3.5333 (0.4125)	0.0410 (0.9902)	3.5294 (0.3650)	3.4667 (0.4239)	0.1340 (0.9002)
Physical demand	1.6471 (0.2564)	1.7333 (0.3003)	1.2850 (0.2419)	1.8824 (0.3195)	1.9333 (0.3581)	0.1470 (0.9000)
Temporal demand	3.1176 (0.2829)	4.3333 (0.4216)	-0.7690 (0.4555)	2.9412 (0.3973)	3.3333 (0.4328)	-0.6370 (0.5355)
Performance	3.4706 (0.4298)	4.2000 (0.4701)	-0.8920 (0.3890)	3.4118 (0.3744)	3.6000 (0.4660)	-0.3100 (0.7646)
Effort	3.6471 (0.3529)	3.0667 (0.3157)	0.1850 (0.8900)	3.0000 (0.2844)	2.8667 (0.3362)	0.3320 (0.7722)
Frustration	3.2353 (0.4580)	2.6000 (0.4000)	-0.2460 (0.8172)	2.8824 (0.4191)	3.0000 (0.4472)	-0.2740 (0.8033)
N	17	15		17	15	
Experience with LLMs						
Mental demand	3.6667 (0.5774)	3.5652 (0.3004)	0.2150 (0.8483)	3.7778 (0.4339)	3.3913 (0.3428)	0.5540 (0.6097)
Physical demand	1.8889 (0.4547)	1.6087 (0.2060)	0.4290 (0.7300)	2.0000 (0.4410)	1.8696 (0.2833)	0.6050 (0.5701)
Temporal demand	4.2222 (0.5212)	3.4783 (0.3072)	1.1540 (0.2654)	3.3333 (0.5270)	3.0435 (0.3527)	0.3640 (0.7281)
Performance	4.4444 (0.6894)	3.5652 (0.3491)	1.1340 (0.2702)	4.5556 (0.6035)	3.0870 (0.2943)	2.1060* (0.0357)
Effort	3.4444 (0.3379)	3.3478 (0.3119)	0.0220 (0.9811)	3.1111 (0.4547)	2.8696 (0.2460)	0.4760 (0.6636)
Frustration	3.2222 (0.6407)	2.8261 (0.3537)	0.4980 (0.6218)	4.0000 (0.6009)	2.5217 (0.3135)	1.9740 (0.0512)
N	9	23		9	23	

Notes. Mean with standard error for the NASA Task Load Index [1] for two groups (high and low) of experience each in searching for data, working with data, and using LLMs, z-statistic and exact p-value for Wilcoxon rank-sum test performed between two groups, number of observations in each experience group, NASA Task Load Index scales range from 1 to 7, * $p < 0.05$.

References

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